Ethics, Privacy, and Security in Data Science and AI (CST4235)

Week 4

(15 Credits)

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Bias and Fairness

Today's Agenda

- Introduction to bias and motivation
- Taxonomy types of common biases
- Mitigate biases in the model/ dataset
- Practical implementation of quantification of Bias and Discrimination in Supervised Learning



What is in this Image?

Watermelon

Watermelon slices

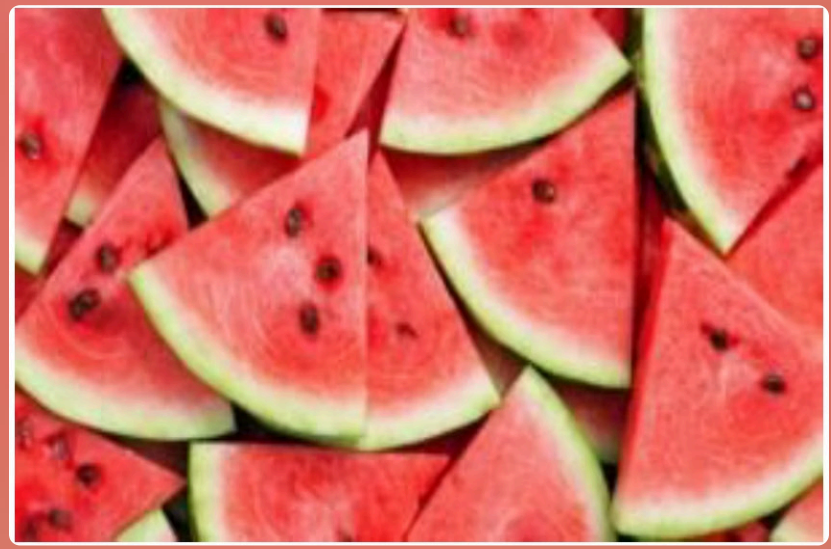
Watermelon with seeds

Juicy watermelon

Layers of watermelon

Watermelon slices next to each other

Did anyone say **RED WATERMELON?**



What is in this image?

It is still a watermelon, just

YELLOW.

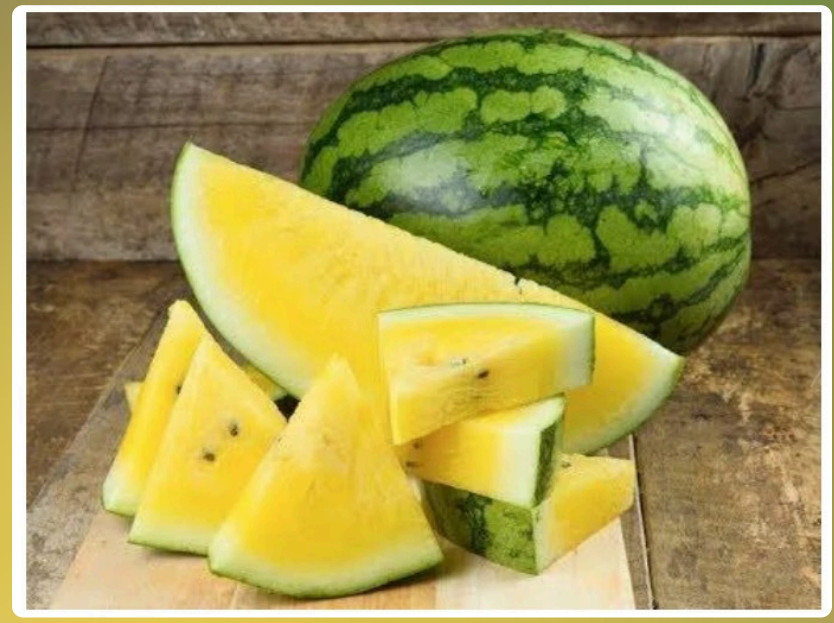
But an AI trained only on red watermelons might not recognise it or might label it incorrectly.

That is how bias happens in AI.

If a dataset contains only red watermelons, the model learns that “red” means “watermelon.”

This is **data bias** – when the system’s understanding is limited by what it has seen.

Bias begins where diversity ends



Labeling, Prototyping, and Stereotyping

Our brains label, AI copies

Our brains constantly simplify information.

We see thousands of colours, faces, and shapes – but we give them labels like “tree,” “car,” “student,” or “leader.”

It helps us make quick sense of the world – and AI does exactly the same.

Prototyping - the “default image” in our minds
A *prototype* is our mental shortcut: When someone says “bird,” most people imagine a small, flying creature – not a penguin or ostrich.

When we see a yellow watermelon or a penguin that doesn’t fly - they surprise our mental models.

Stereotypes - When particular **labels** and **features** get linked too tightly
(e.g., leader = male, nurse = female, beauty = fair skin) - they distort our judgments.

AI systems trained on such biased patterns **replicate** these same stereotypes – whether in hiring, image tagging, or voice recognition.

Cognitive Bias and Gender Stereotypes in AI and Humans

A man and his son are in a terrible accident and are rushed to the hospital in critical condition.
The doctor looks at the boy and exclaims:

"I can't operate on this boy — he's my son!"

How could this be?

Most people struggle because their brain automatically assumes the doctor is a man.

That pause, that confusion – *that's bias in action.*

Our **mental prototype** for "doctor" is often male – shaped by culture, media, and repetition.

We unconsciously link **roles** with **genders** – "doctor = male", "nurse = female", "leader = man".

AI systems trained on biased datasets pick up the *same associations*.
For example, image-tagging algorithms have labelled "women cooking" and "men at work" simply because of biased training data.

Bias is not just a technical flaw; it's a reflection of us.



Learning from Text – When AI Learns Our Words, It Learns Our World

Gordon & Van Durme (2013) analysed millions of words from online text to see what AI models might "learn" about the world.

The table shows word frequency – how often words appear in huge text datasets.

Notice something?

Violent words like **"murdered"** appear far more frequently than gentle words like **"hugged"** or **"breathed."**

What This Means for AI?

If the internet talks more about violence than kindness, AI starts to associate humans with aggression.

This is **statistical bias** – not because the model "hates," but because it reflects what it sees most.

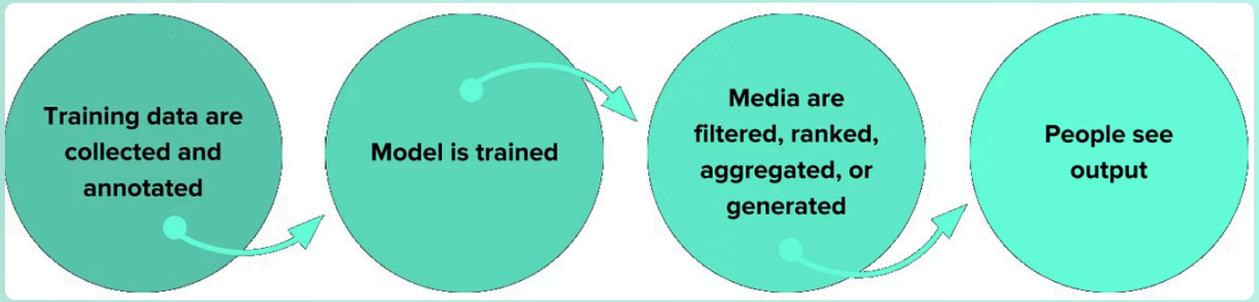
AI doesn't understand meaning — it counts meaning.

Word	Frequency in corpus
"spoke"	11,577,917
"laughed"	3,904,519
"murdered"	2,834,529
"inhaled"	984,613
"breathed"	725,034
"hugged"	610,040
"blinked"	390,692
"exhale"	168,985

What is Bias?

Definition (Ntoutsi et al., 2020):

Inclination or prejudice of a decision made by an AI system which is for or against one person or group, especially in a way considered to be unfair.



Standard ML Pipeline

Bias doesn't wait for the output — it sneaks in at every step

Training data → Model → Output → People

1) Data Collection & Annotation:
Where bias *is born.* If data overrepresents one group or culture – the model learns a biased view of the world.

2) Model Training:
The model doesn't know right from wrong – it just copies patterns.
If biased data go in, unfair decisions come out.

3) Filtering & Ranking:
Now the model decides what you see online – news, jobs, even people.
Bias here shapes visibility and opportunity.

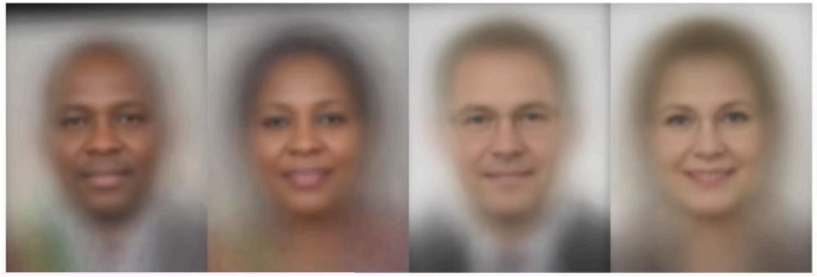
4) People See the Output:
We trust what AI shows us, often without questioning it.
That's how bias turns from a *technical flaw* into a *social impact.*

Bias in Facial Detection

When machines see some faces better than others

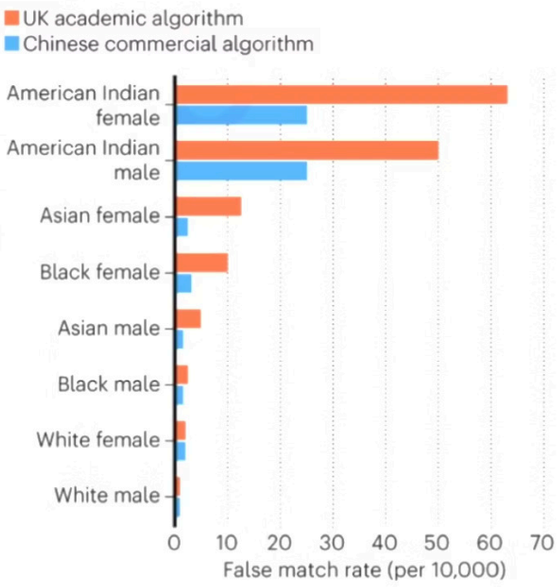
Independent Study I

Gender Classifier	Darker Male	Darker Female	Lighter Male	Lighter Female	Largest Gap
Microsoft	94.0%	79.2%	100%	98.3%	20.8%
FACE++	99.3%	65.5%	99.2%	94.0%	33.8%
IBM	88.0%	65.3%	99.7%	92.9%	34.4%



Buolamwini/Gebru *AAAI/AIES* 2018.
Nature 2019.

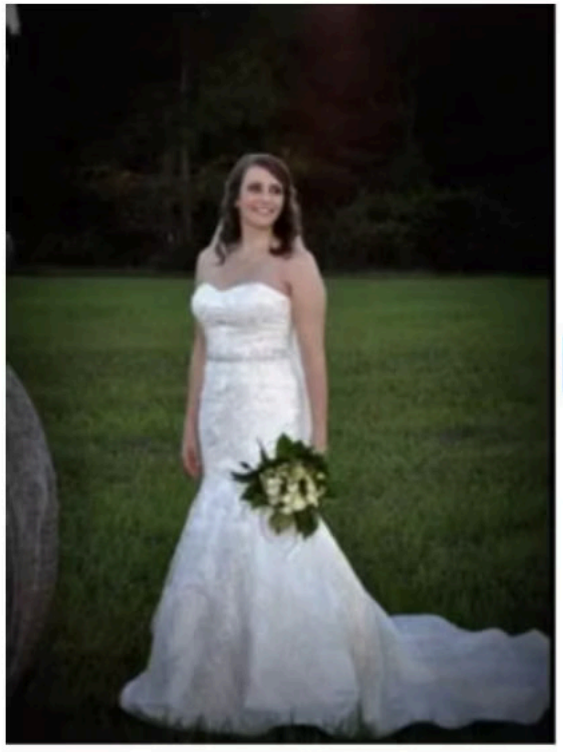
Independent Study II



- Independent tests (Buolamwini & Gebru, 2018; *Nature*, 2019) showed major tech companies' gender classification systems were **up to 34% less accurate** for darker-skinned women compared to lighter-skinned men.
- In another global comparison, **false match rates** were highest for *American Indian and Black females* – proving that even large-scale algorithms "see" people unevenly.

AI systems used for hiring, surveillance, or law enforcement may unfairly target or exclude certain groups – **not because they're designed to discriminate, but because their training data didn't represent everyone equally.**

Bias in Image Classification



Ground Truth: Bride

CNN

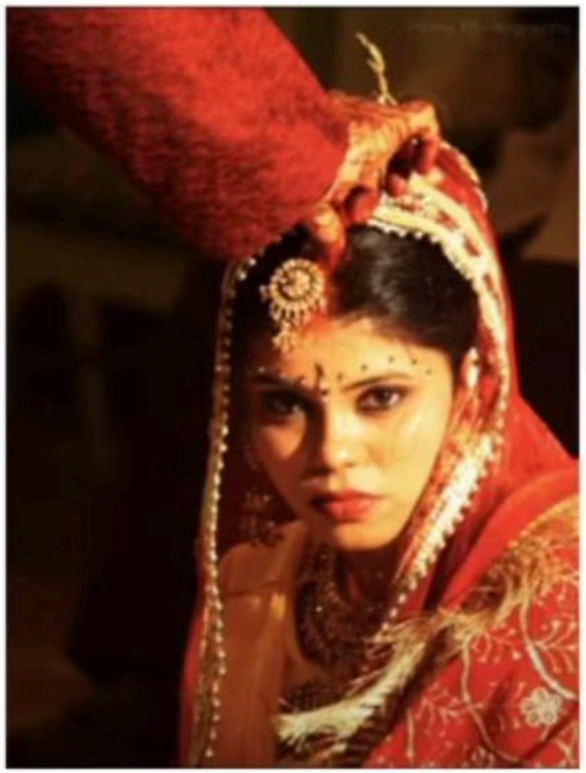
CNN for image classification.

Predicted Classes

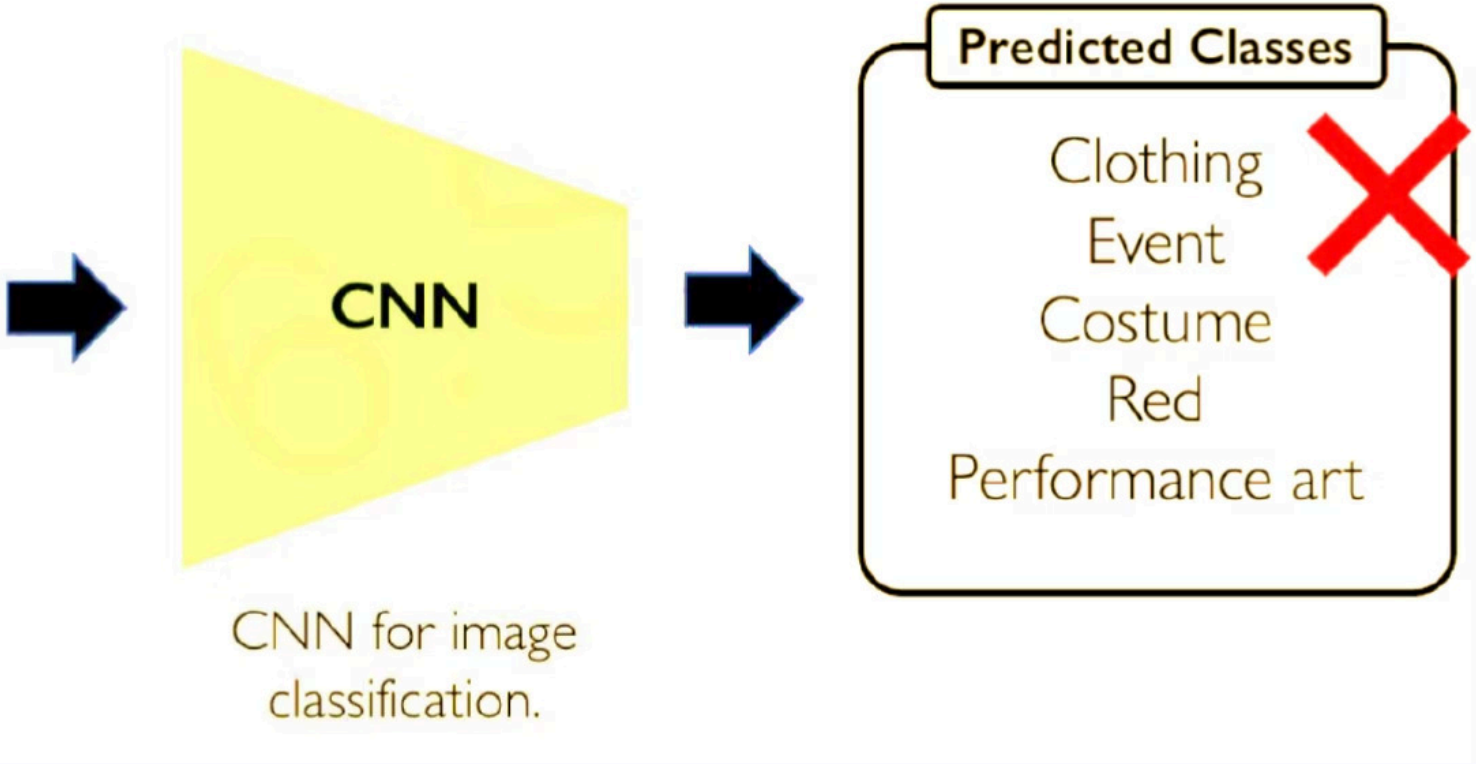
Bride
Dress
Ceremony
Woman
Wedding

Bias in Image Classification

A wedding to us, a 'costume' to AI.

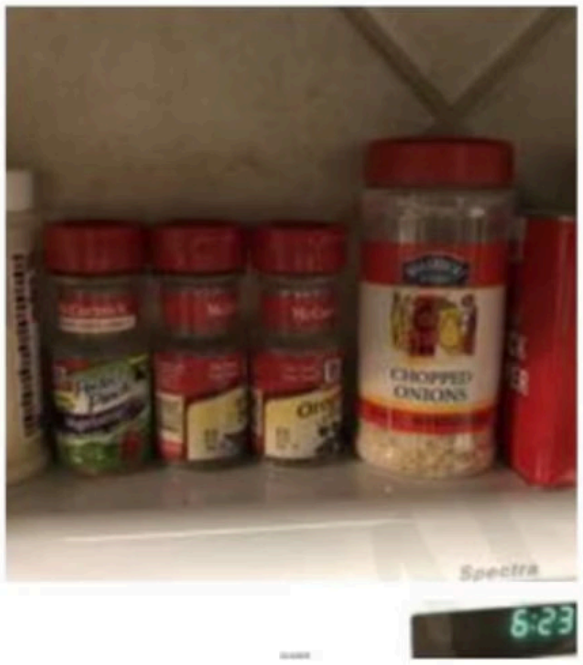


Ground Truth: Bride



CNN for image classification.

Bias in Object Recognition



Ground Truth: Spices

CNN

CNN for object recognition.

Predicted Objects

Seasoning

Spice

Spice rack

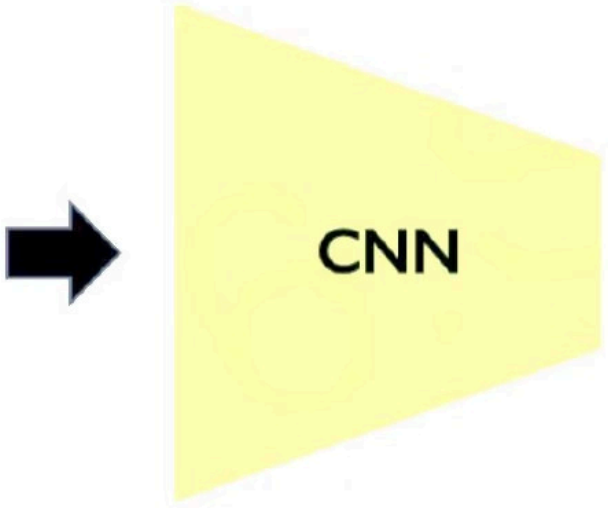
Ingredient

Bias in Object Recognition

Bias isn't just about faces — it's in the kitchen too.



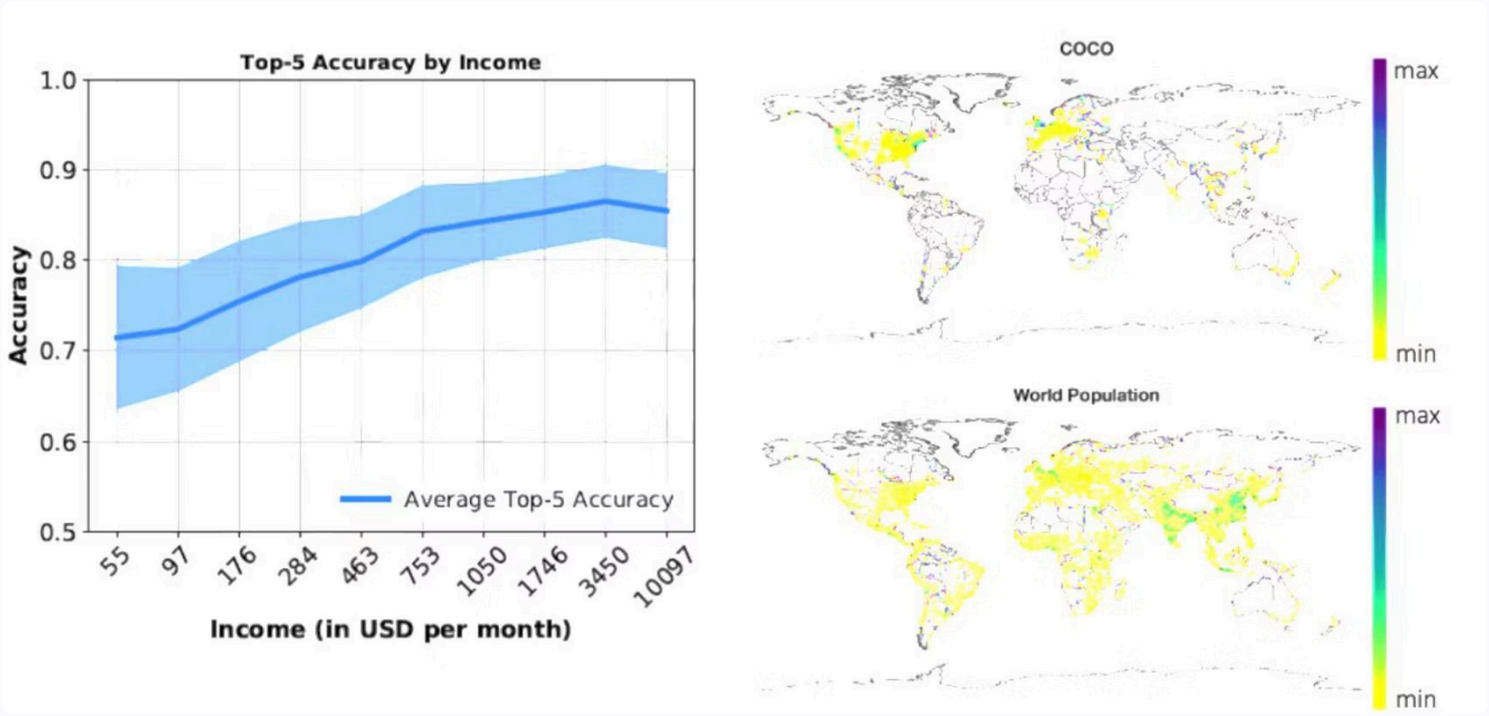
Ground Truth: Spices



CNN for object recognition.



Bias Correlation with Income and Geography



- As **income increases**, AI accuracy also **increases**
- AI systems perform better in wealthy countries and worse in low-income regions.
- The COCO dataset (used to train image recognition AI) mostly includes data from North America and Europe.
- But the world population is concentrated in Asia and Africa – areas that are underrepresented in AI training data.
- AI "learns" more about some parts of the world than others.
  So, it may recognise a toaster in New York, but not a cooking pot in Nairobi.

Bias in Voice AI systems

When your voice decides if you're understood.

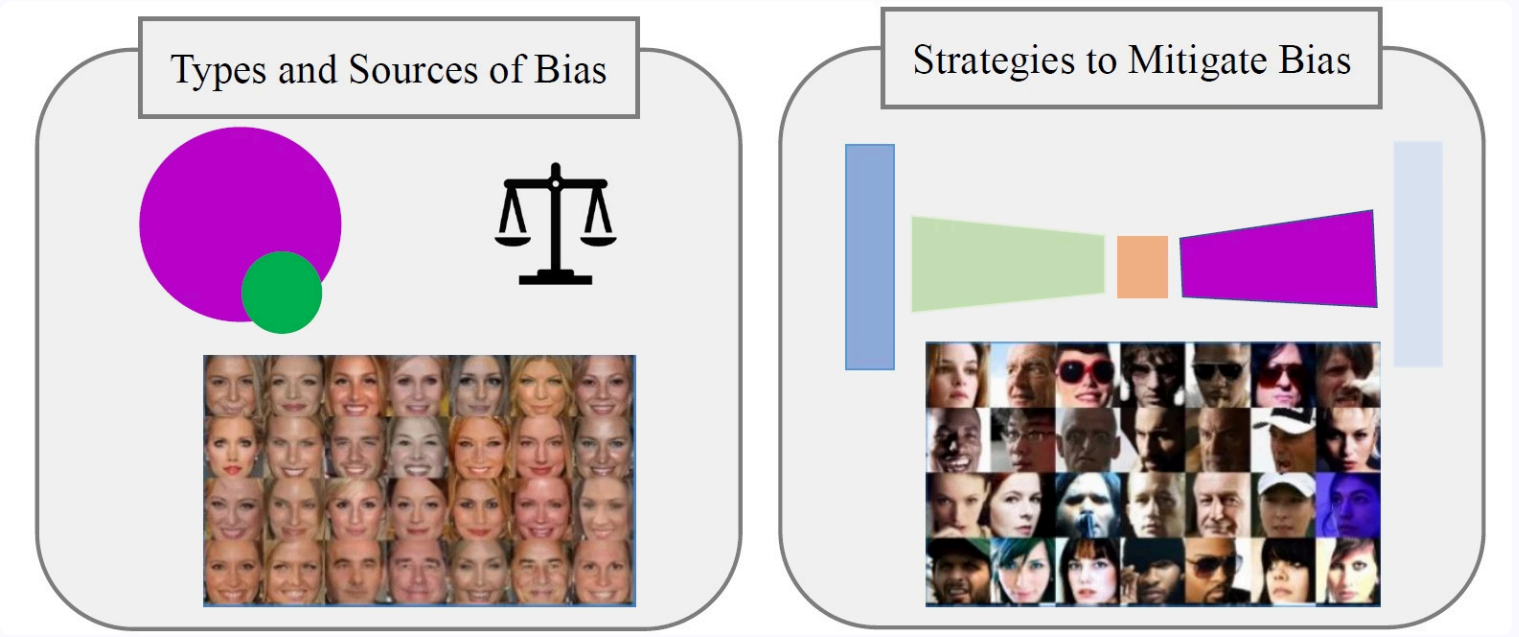
Voice assistants like **Alexa, Siri, or Google Assistant** are trained mostly on certain accents, tones, and speech patterns.

This means they recognise voices from some groups better than others.

Example: A Voice AI may easily understand someone with an American or British accent, but struggle with a non-native English speaker or even with background noise.

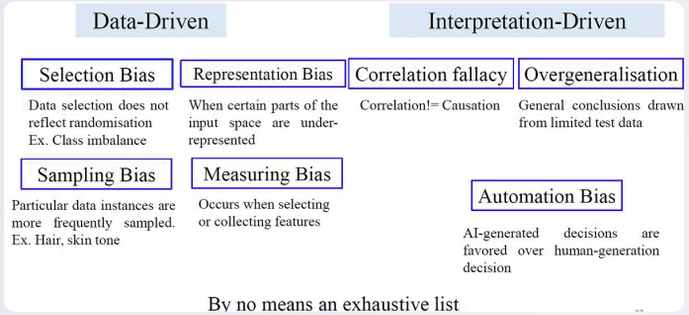


Understanding and Mitigation Algorithmic Bias



When one type of face dominates the data, the AI starts to “believe” that is what everyone looks like – creating invisible walls of exclusion.

By widening the lens with diverse data and balanced learning, AI begins to see the real world – fairer, broader, and more human.



Taxonomy of Common Biases

Data-Driven Biases

Where it all begins: in the data itself. Before a single model runs, bias quietly sneaks in.

- **Selection Bias:** When your data forgets diversity – the model never learns balance.
- **Sampling Bias:** Some features get the spotlight; others stay invisible.
- **Representation Bias:** The world's variety reduced to a narrow lens.
- **Measuring Bias:** Even your sensors can lie – garbage in, bias out.

Interpretation-Driven Biases

Now bias wears a new mask – hidden in how we read or trust AI.

- **Correlation Fallacy:** Just because two things move together doesn't mean they're linked.
- **Overgeneralisation:** Small dataset, big (wrong) conclusions.
- **Automation Bias:** When humans start believing the machine knows best – even when it doesn't.

Correlation Fallacy

When it is a coincidence, not a cause.

Ice Cream Sales vs. Drowning Rates

Both go up in summer – but eating ice cream does not cause drowning.

The real cause is hot weather, which increases both swimming and ice cream consumption.

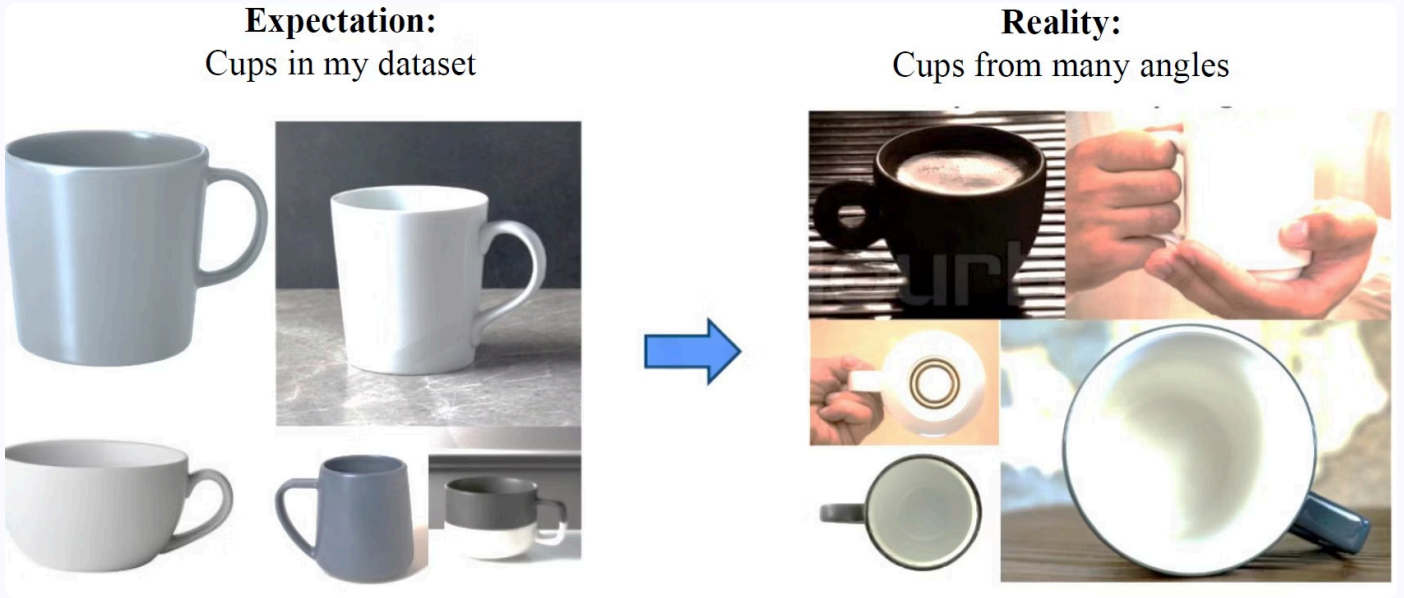
Police Presence vs. Crime Rate

Cities with more police also report more crimes – not because police cause crime, but because high-crime areas require more policing.



Bias from Overgeneralisation

When the model thinks one view tells the whole story.



Expectation: Cups in my dataset – clean, centred, well-lit.

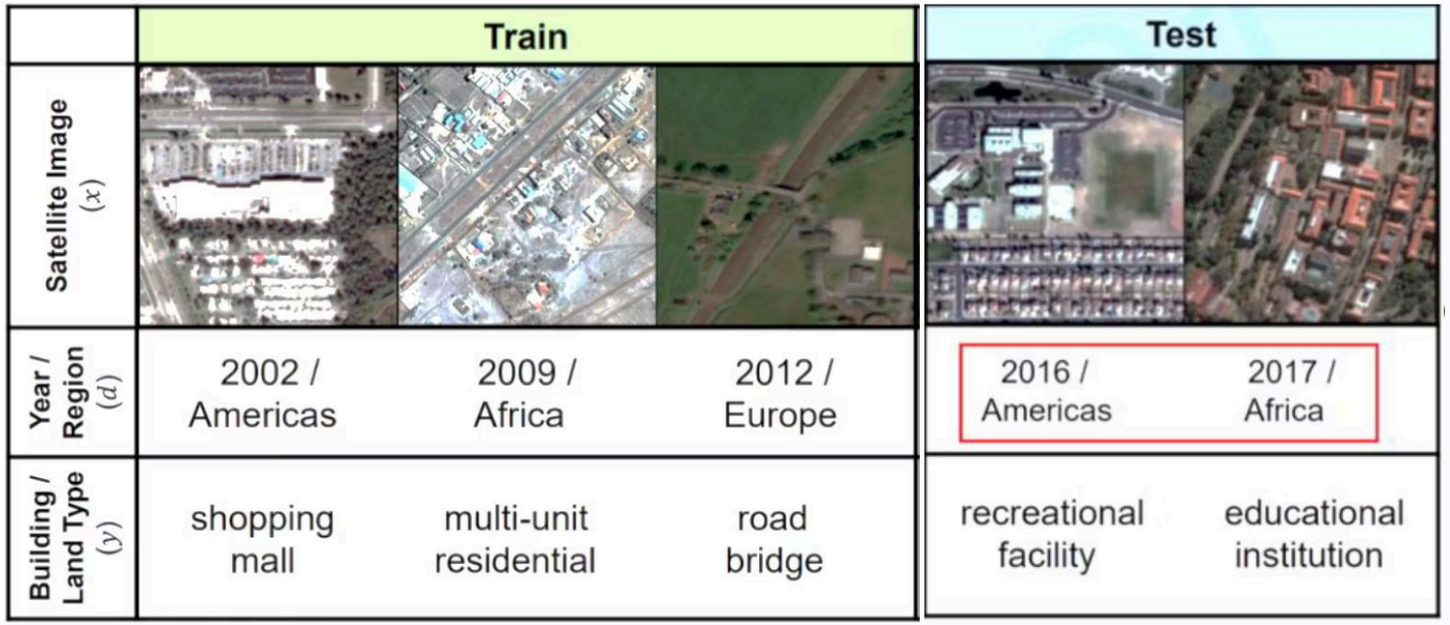
Reality: Cups in the real world – different colours, angles, lighting, and contexts.

When training and real-world data distributions differ, the model's confidence stays high – but its accuracy collapses.

Overgeneralisation = assuming "what I have seen" means "I have seen it all."

Dataset with Distribution Shifts

When yesterday's data fails to understand today's world.



Building/Land Classification using satellite images

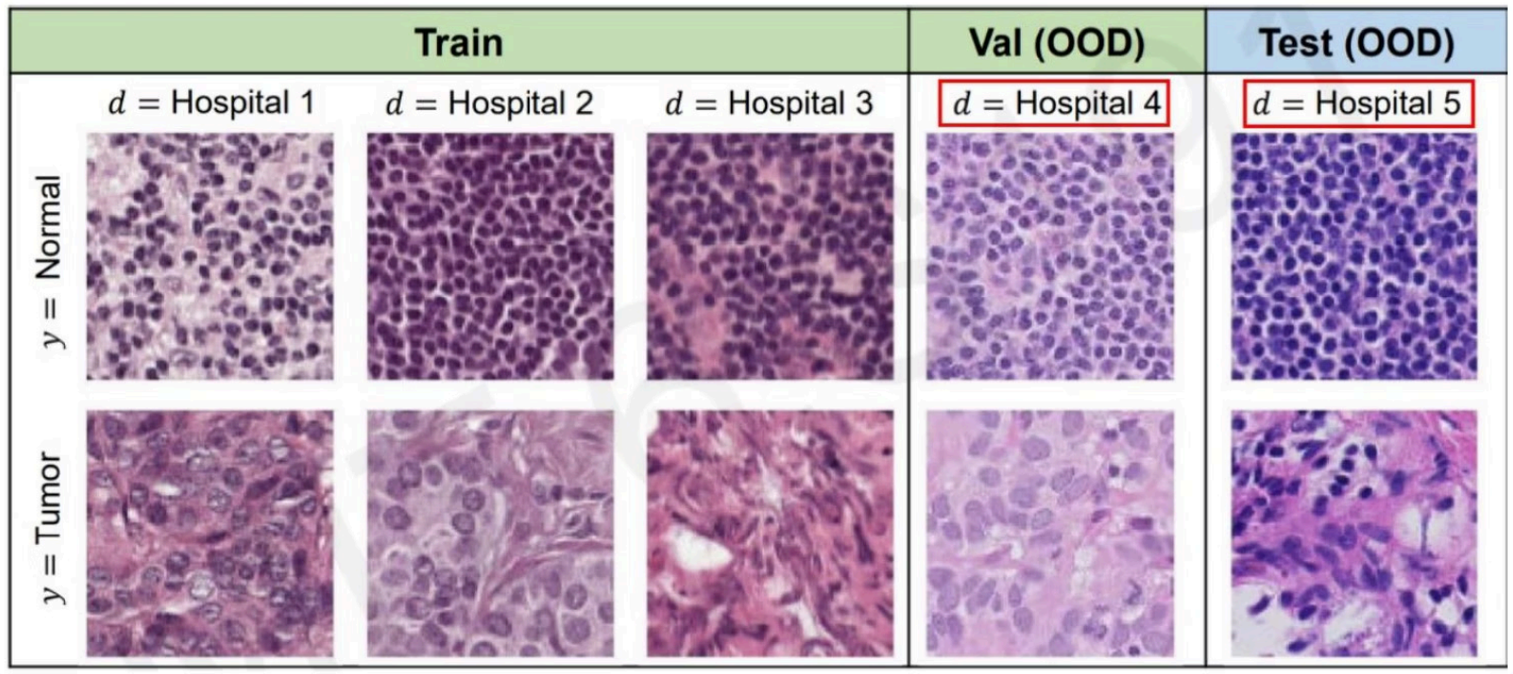
The model was trained on data from different years and regions – 2002 (Americas), 2009 (Africa), 2012 (Europe) **but tested on new data — 2016 (Americas), 2017 (Africa)**

Result: The model struggles to recognise new patterns – lighting, land use, and architecture all changed. Accuracy drops because the world evolved, but the model didn't.

Distribution shift **= when the training and test data come from different worlds (time, region, or context).**

Dataset with Distribution Shifts

Same disease, different hospitals — the model gets confused.



Scenario: Disease classification from histopathology images

The model was trained on data from **Hospital 1-3**, but tested on data from **Hospital 4-5**.

Problem: Different hospitals use different staining techniques, scanners, lighting, and magnifications. So, even though the cells look “similar” to us, the AI sees them as different.

Result: The model misclassifies healthy vs tumour cells – not because it lacks skill, but because the data distribution changed.

Historical Bias

AI doesn't just learn from data — it inherits society's past.

Bias that already exists **before any data collection** or modelling

Example:
Search for *"CEO"*:

For years, the results showed **mostly men**,
because the real-world data reflected that – in 2018, only 5% of Fortune 500 CEOs were women.

What Google did:
Google adjusted its image search algorithm to include **more women** – not to distort reality,
but to challenge historical imbalance and promote fairer representation.

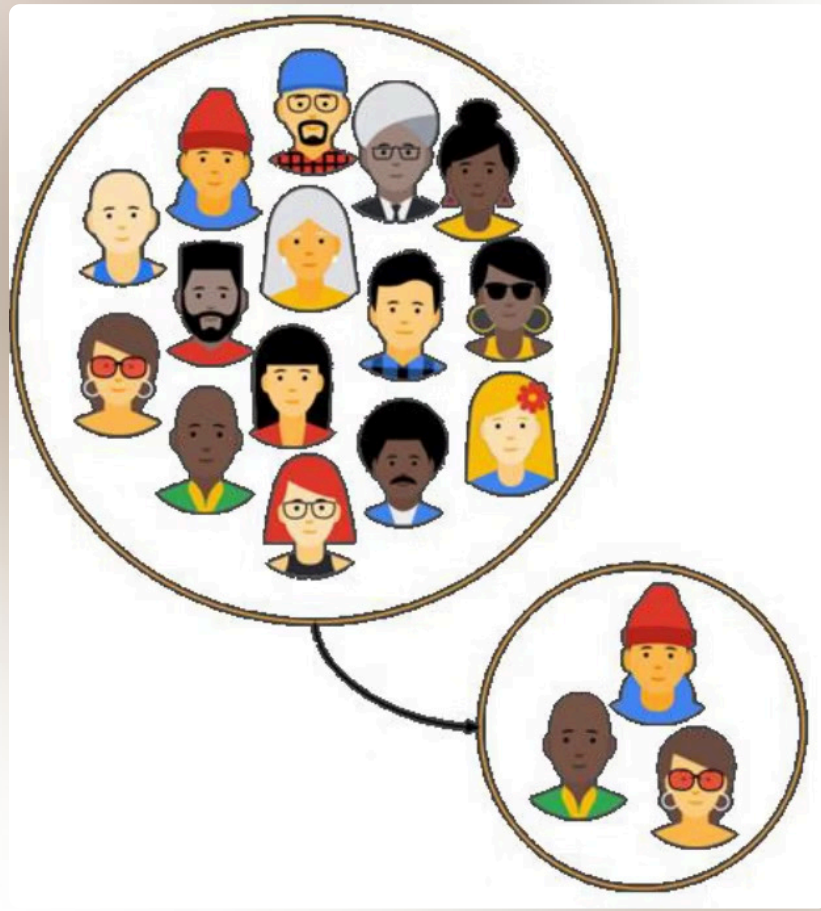


Representation Bias

Even if all groups appear in your dataset, some may be shown less frequently or in less positive contexts, creating a biased view of reality.

Example: In an image dataset, there might be photos of both men and women but if most men are shown in engineering roles and most women are shown in support roles,
the AI learns that “engineer = man” and “assistant = woman.”

Representation bias isn’t about missing data, it is about imbalanced visibility and biased portrayal.

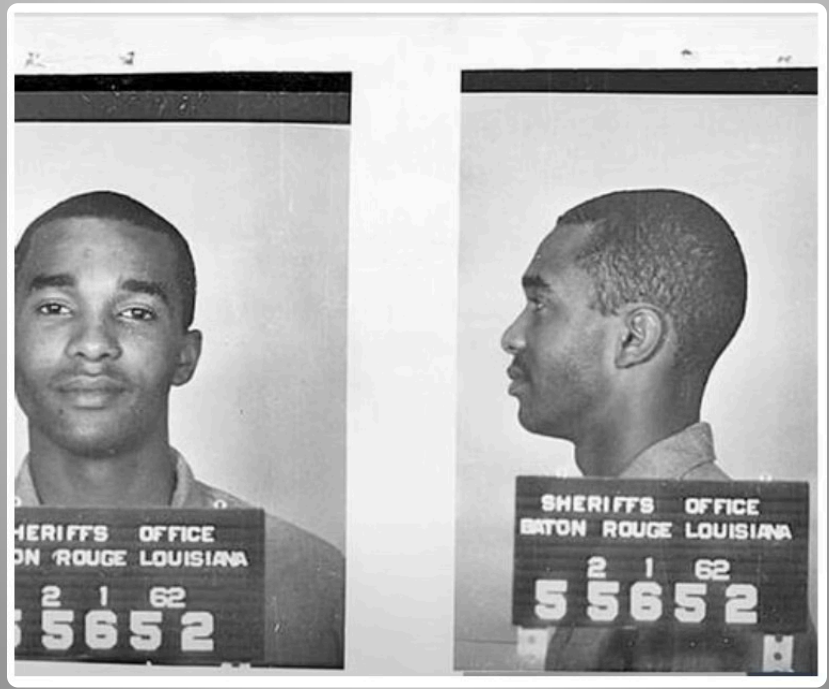


Measurement Bias

When what we measure does not truly reflect reality.

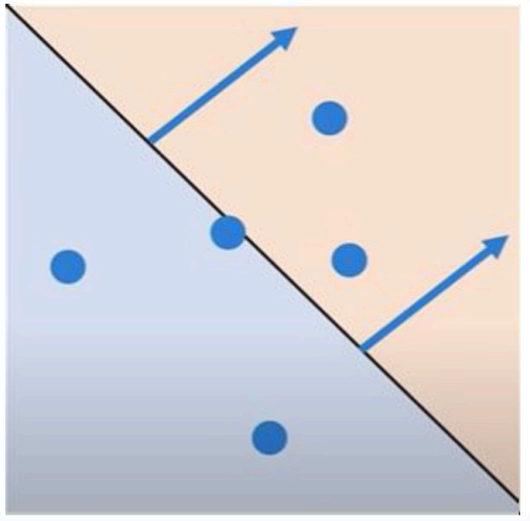
Example 1: Using **arrest rates** as a measure of **crime rates** assumes all groups are policed equally but in reality, some communities are over-policed, inflating their “crime” statistics.

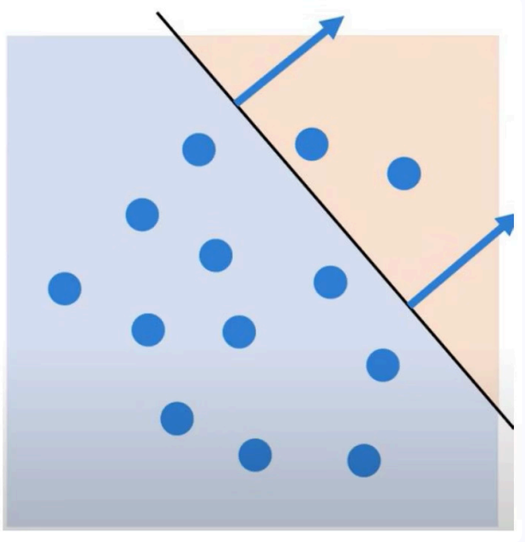
Example 2: The **COMPAS** system used prior arrests (including friends’ or family arrests) to predict a person’s **likelihood of reoffending (recidivism)**

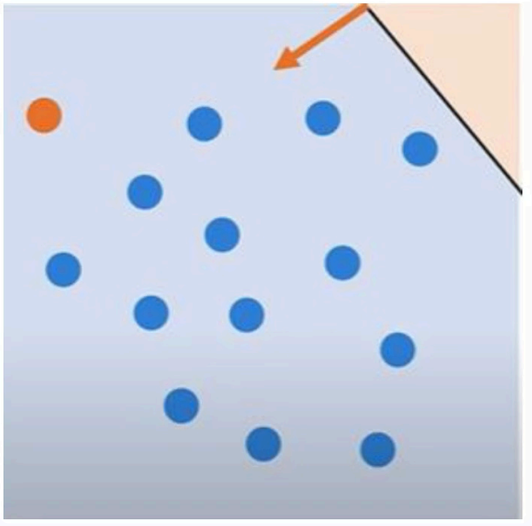


Effect of Class Imbalance

When one class is huge (say 20 normal cases for every 1 rare case),
the model starts thinking the majority is “normal” and stops paying attention to the rare ones.





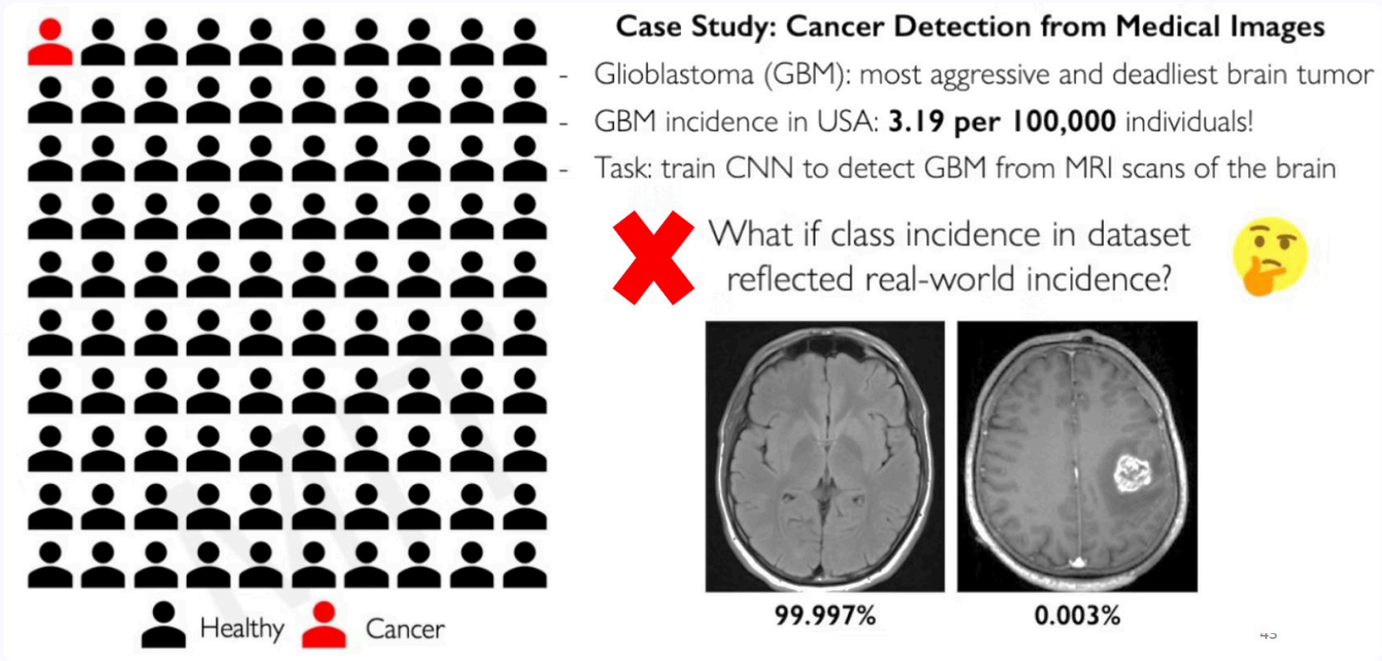


- Blue = minority class (only a few examples)
- Orange = majority class (lots of examples)
- The black line = model’s boundary between them

Because the orange class dominates, the model keeps adjusting the line towards the blue side trying to please the majority.

Result: The rare class gets ignored or misclassified.

The Danger of Class Imbalance (Cancer Detection Case Study)



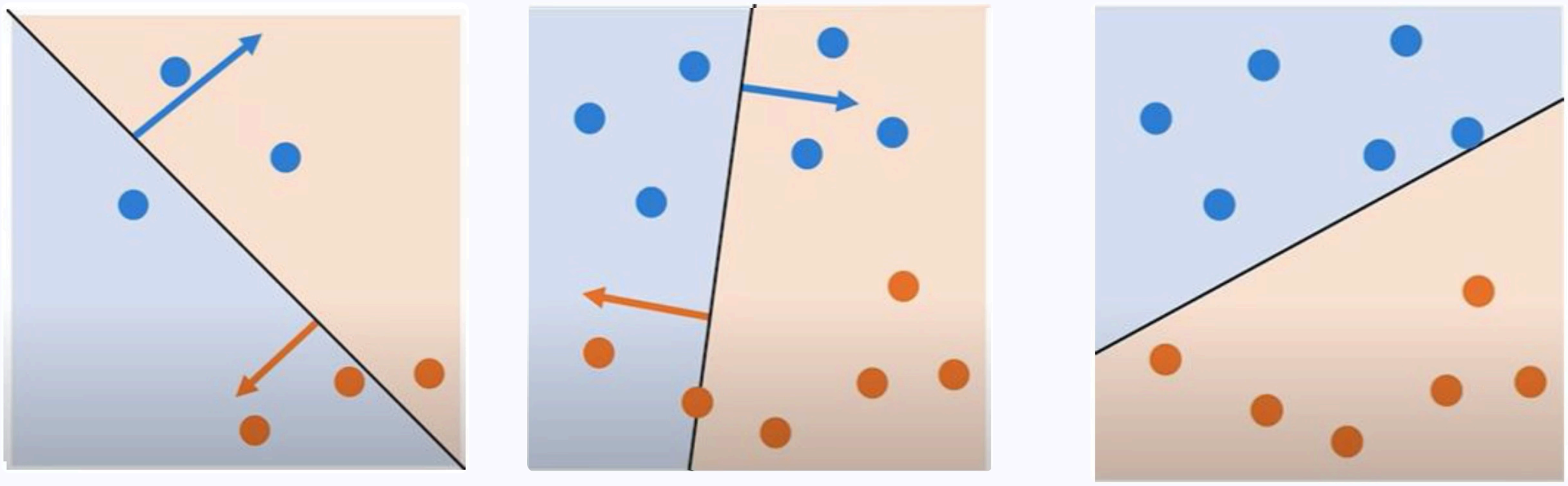
- You are training a model (like a CNN) to detect **Glioblastoma (GBM)**, a rare but deadly brain tumour.
- In real life, GBM only affects about **3 in every 100,000 people** – extremely rare.
- The image of many black icons (healthy) and one red icon (cancer) shows that the dataset is hugely imbalanced, almost all "healthy."

Result: The model will learn to always predict "healthy" – because that gives it almost 100% accuracy. But in reality, it completely fails to detect cancer, which is the class we care most about.

If your data reflects reality too literally, your model might be technically accurate but clinically useless.

Learning from Class-Balanced Data (1:1 Ratio)

When training a model, we can make sure that each batch (small group of data) has the same number of examples from both classes – for example, 1 cancer and 1 healthy image.



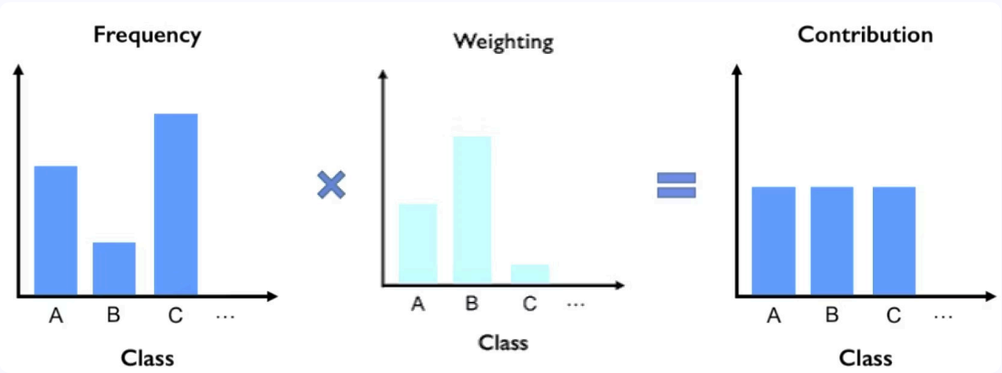
- The black line is the model's decision boundary (it tries to separate the two classes).
- At the start, it is random not accurate yet.
- As the model learns, it updates step by step (incremental updates).
- Because each batch has a 1:1 ratio, the model gets equal information from both sides.
- Slowly, the boundary moves to the correct position not biased to one side.

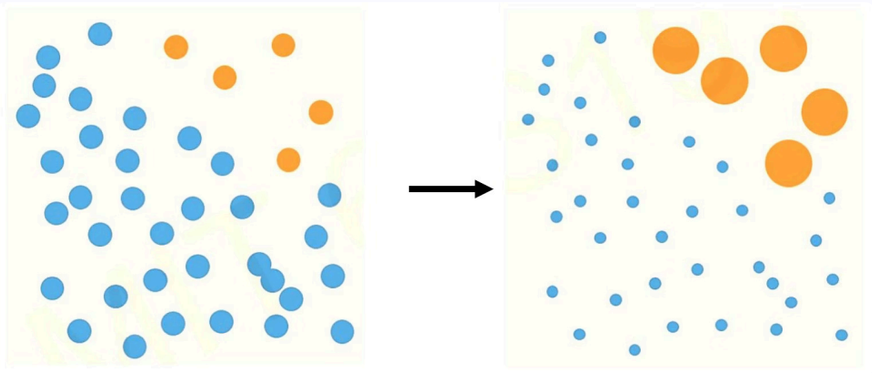
If the model only sees healthy patients most of the time, it will think everyone is healthy. But if we show it an equal number of healthy and cancer examples in each batch, it starts learning to tell them apart correctly.

Learning from Class-Balanced Data: Example Weighting

When we cannot collect more data for rare classes, we can give more importance (weight) to those rare examples during training.

This makes the model pay extra attention to them as if they appeared more often.



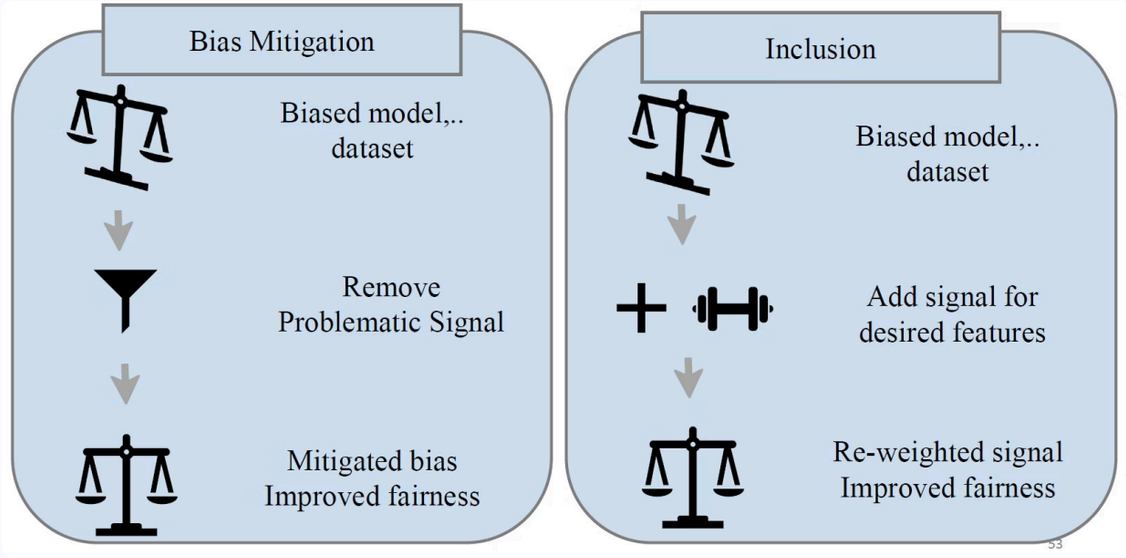


- **Left (Bar chart):**
  - Shows that some classes (like C) appear more often than others (like B).
  - To fix this, we assign higher weights to rare classes
  - Result → all classes contribute equally during learning.
- **Right (Dots illustration):**
  - The size of each dot = how much importance
  - Before weighting → rare (orange) dots are small and few.
  - After weighting → rare dots become bigger and more likely to be used for training.

Is the “Gold Standard” Really Golden?



Making AI Fair: Data and Learning Techniques That Matter



Fix the Bias

Remove what's unfair

- Check for skewed data or hidden patterns
- Remove or reduce signals that cause unfairness (e.g., gender, age)
- Train models again with cleaned data

Result: Less bias, fairer predictions

Add What's Missing

Include what's left out

- Bring in data from multiple, diverse sources
- Add more examples for underrepresented groups
- Ask domain experts to find missing signals

Result: Balanced data, inclusive models

Data really matters: Remove the bad, add the missing, and fairness follows.

How Do We Check If AI Is Fair? (Evaluation)

1. **Disaggregated Evaluation - One Feature at a Time**

- We test fairness by looking at each group separately.
- For example: How well does the model perform for each colour (blue, orange, green)?
- Or for each shape (circle, triangle, square)?
- This helps find bias in single features – like gender or age.

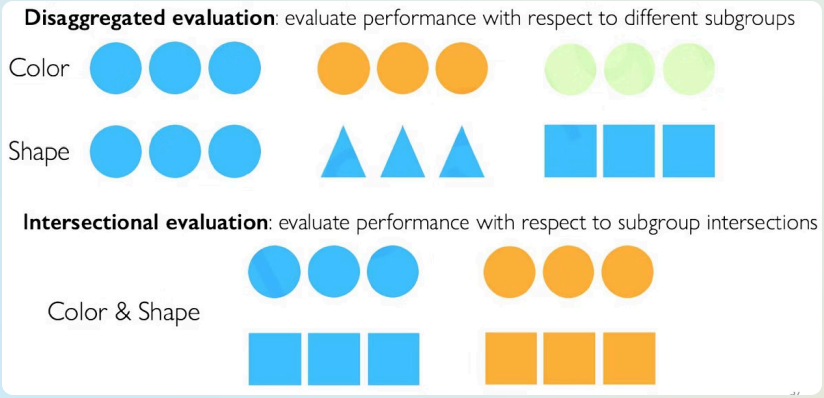
2. Intersectional Evaluation - Combined Features

We test fairness by looking at combinations.

Example: Blue circles, orange squares, green triangles.

This finds hidden bias that appears only when features overlap (like “young women” or “older men”).

Checking one group at a time is good. Checking combinations is even better — that’s where hidden bias hides.



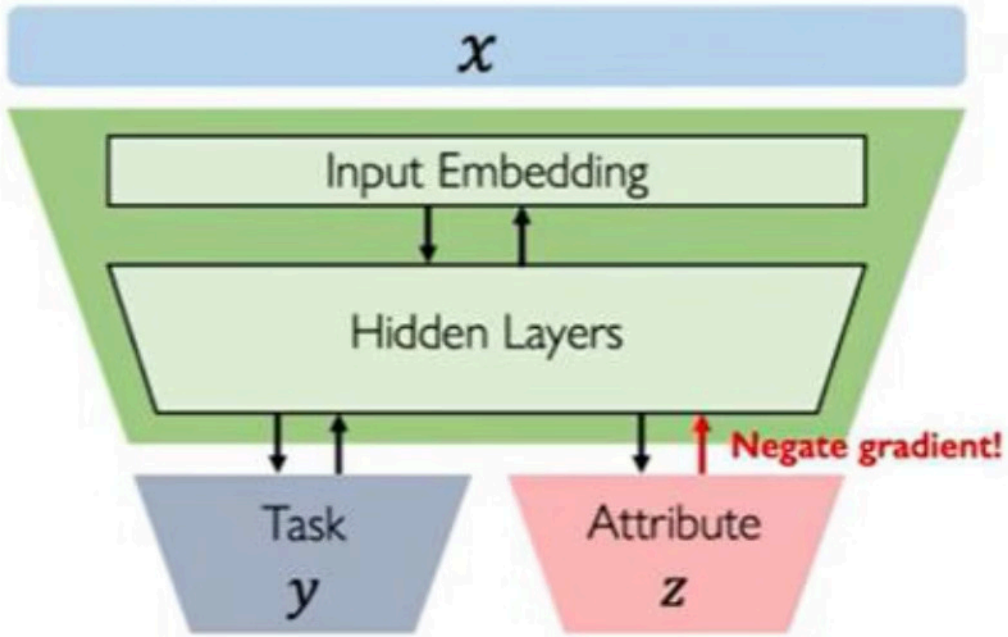
Application To Language Modeling

Task: language model to complete analogies

He is to **she**, as **doctor** is to ?

biased		debiased	
neighbor	similarity	neighbor	similarity
nurse	1.0121	nurse	0.7056
nanny	0.9035	obstetrician	0.6861
fiancée	0.8700	pediatrician	0.6447
maid	0.8674	dentist	0.6367
fiancé	0.8617	surgeon	0.6303
mother	0.8612	physician	0.6254
fiance	0.8611	cardiologist	0.6088
dentist	0.8569	pharmacist	0.6081
woman	0.8564	hospital	0.5969

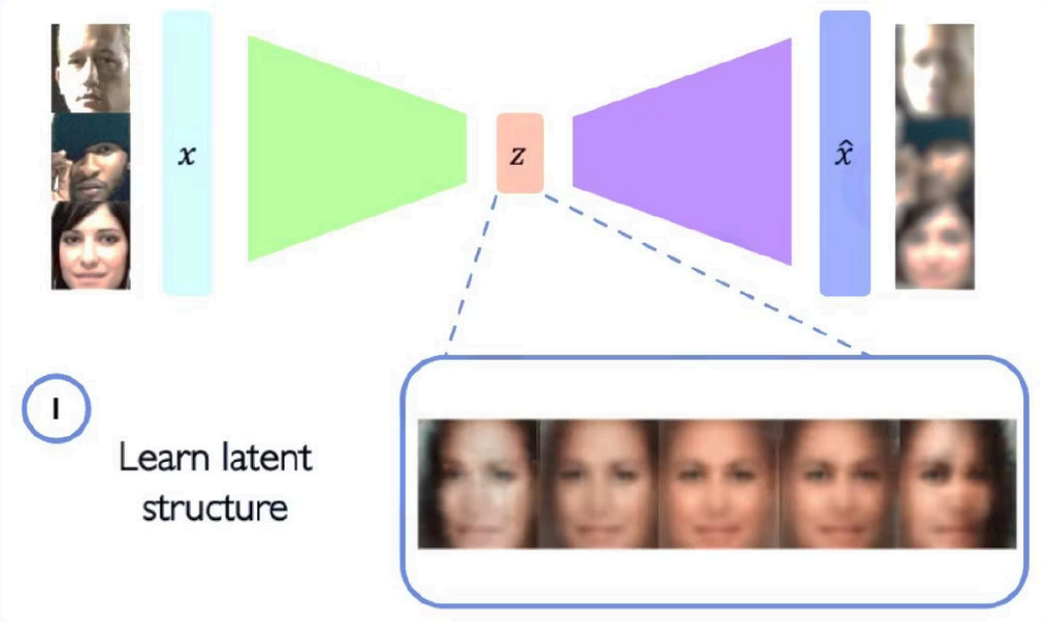
Sensitive attribute: Gender



Jointly predict output label y and sensitive attribute z to remove from decision

Mitigating Bias through Learned Latent Structure

Step 1: The Model Learns Patterns from Faces

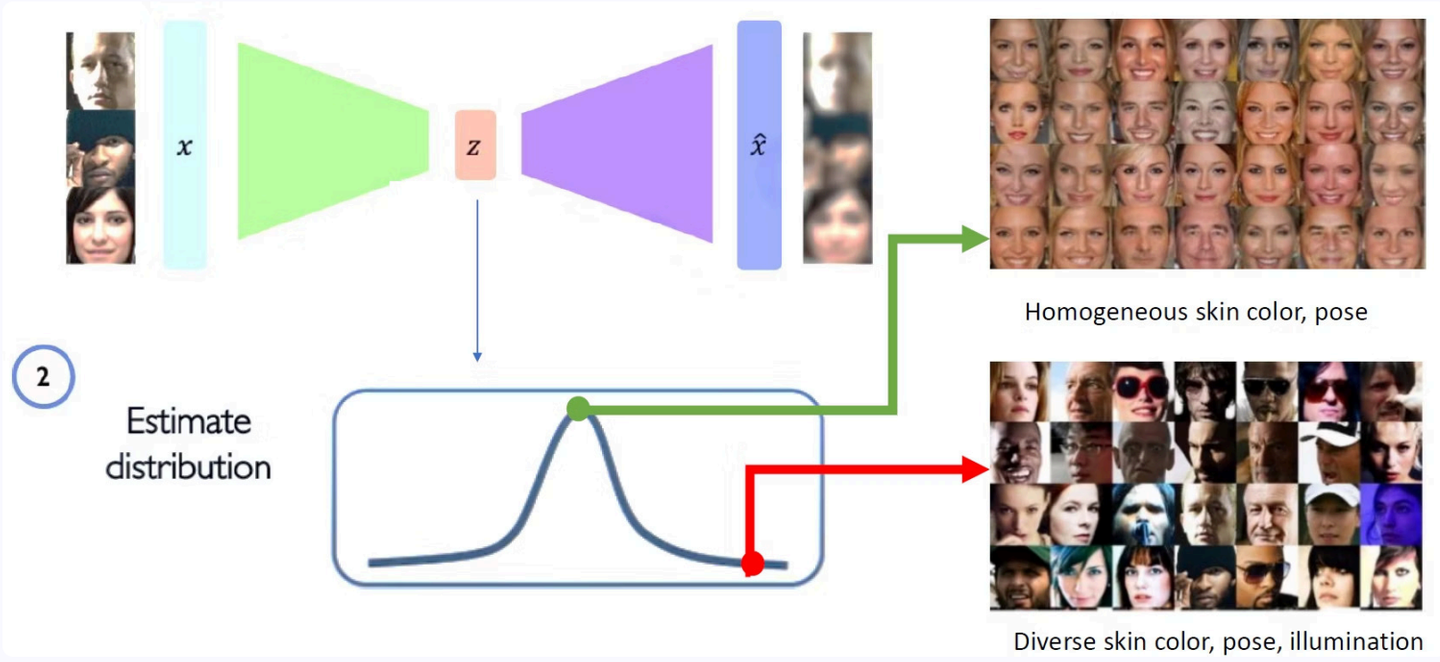


- The pictures on the left (x) are the input images.
- The model looks at these pictures and tries to understand common patterns – such as skin colour, face shape, and lighting.
- These patterns are stored inside the model as *latent structure (z)* – a kind of "summary" of what the faces look like.
- The model then tries to rebuild (recreate) the faces on the right (x̂) using what it has learned.

The model is learning what faces usually look like — but it may already start to pick up bias if some faces are more common than others in the data.

Mitigating Bias through Learned Latent Structure

Step 2: Estimating Bias in What the Model Learned

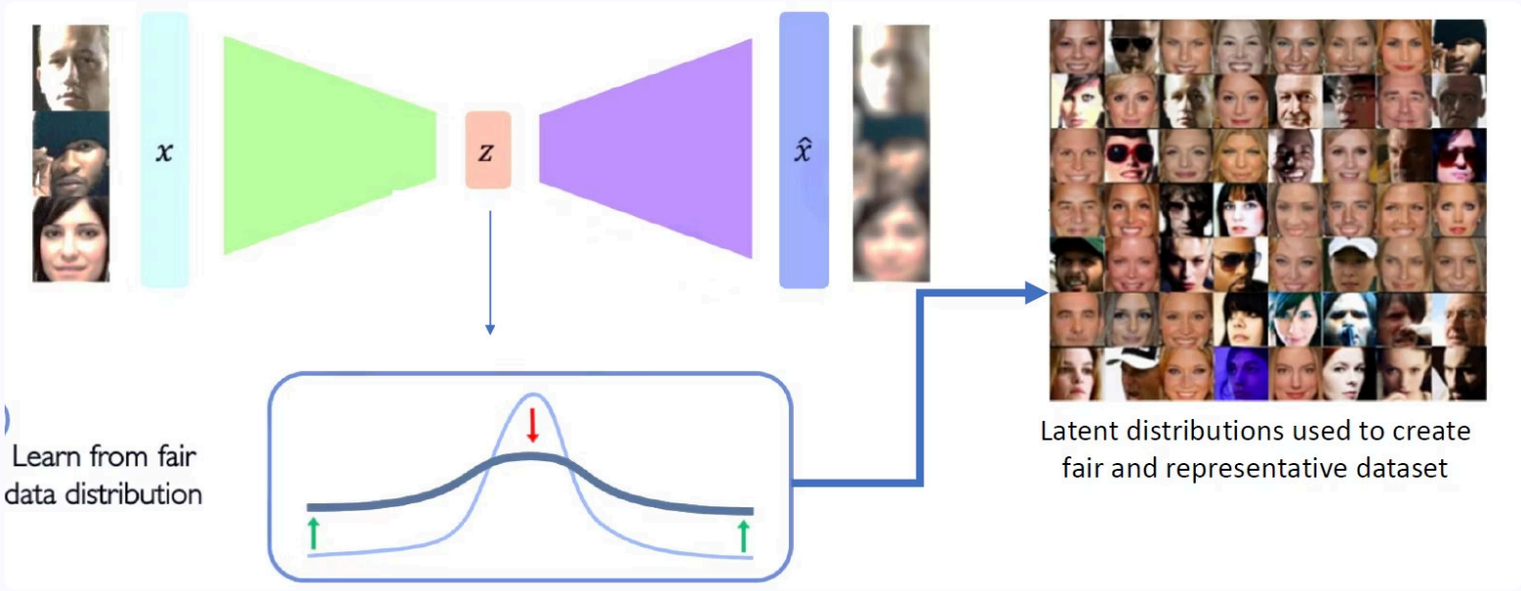


- After learning, the model checks what kind of faces it has seen most often.
- If the model has mostly *similar faces* (e.g., same skin colour, same pose), it means the data is **biased**.
- If the model has *diverse faces* (different skin colours, lighting, and poses), it means the data is **fairer and more balanced**.
- The curve below shows this – one side represents limited variety, the other side shows diversity.

The model estimates how fair or biased the learned patterns are

Mitigating Bias through Learned Latent Structure

Step 3: Learning from a Fair Dataset

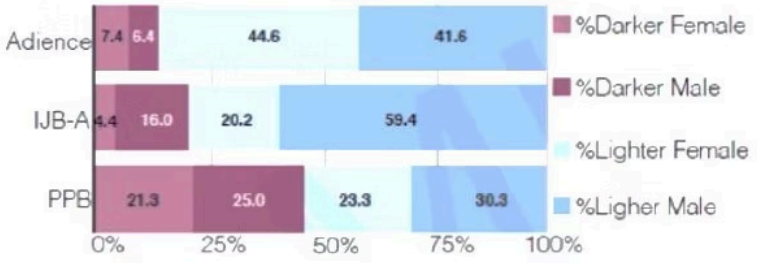


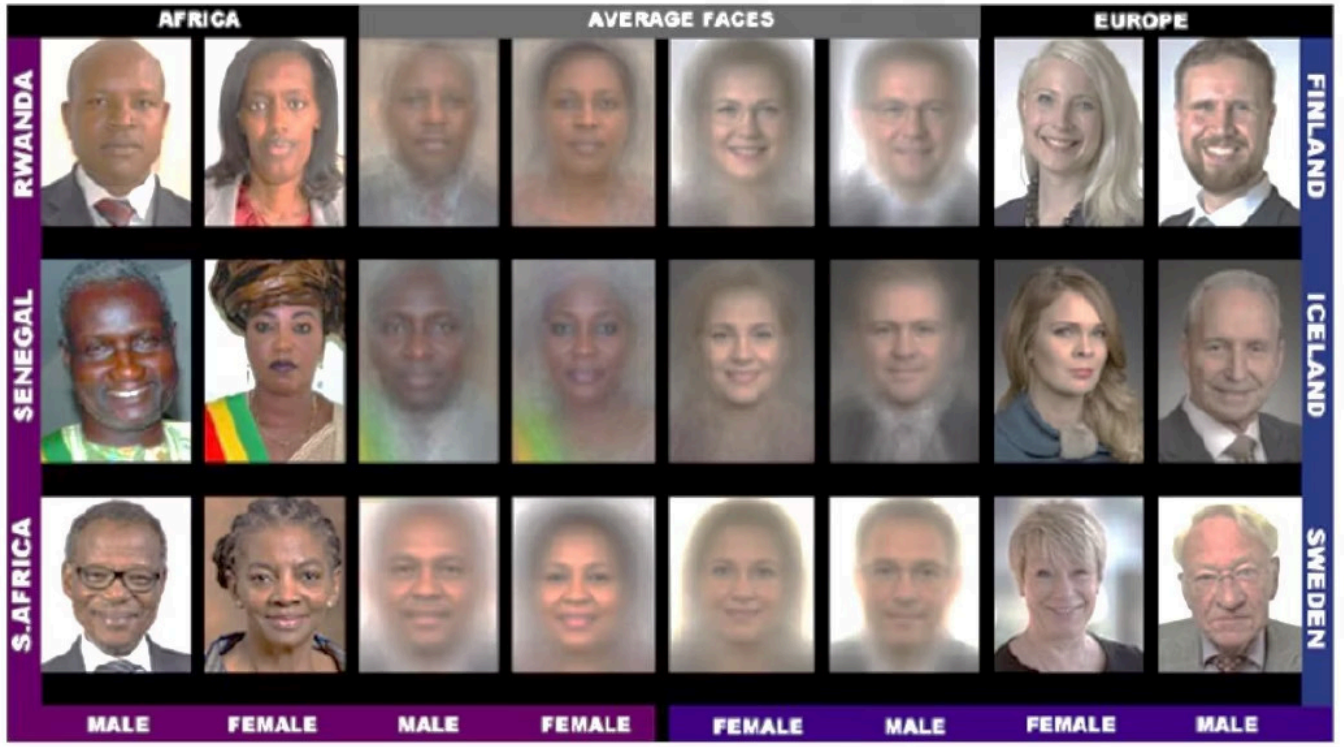
- The model now adjusts its learning to **balance the data**.
- It gives **less weight** to overrepresented faces (too many similar ones).
- It gives **more weight** to underrepresented faces (less common ones).
- This creates a fair and diverse dataset that includes different skin tones, lighting, and poses.
- The model now learns from a more representative and unbiased view of people.

The model rebalances what it has learned to treat all groups fairly.

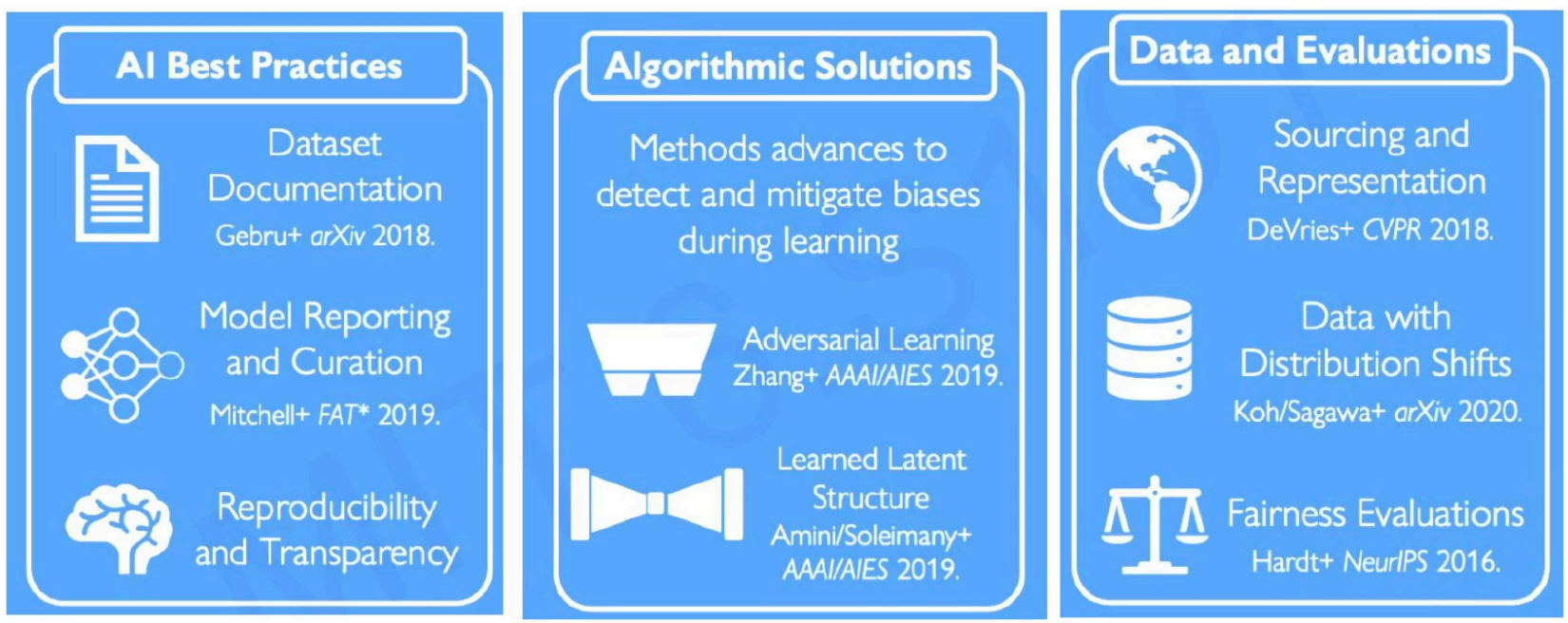
Balanced Dataset for Evaluation

- Pilot Parliaments Benchmark (PPB) dataset
- Evaluation of facial detection algorithms
- Skin tone, male/female binary





AI Fairness: Summary and Future Considerations



1. **AI Best Practices** – Be transparent, document datasets, and report clearly.
2. **Algorithmic Solutions** – Use smarter learning methods that detect and fix bias.
3. **Data and Evaluations** – Use balanced, diverse, and representative datasets (like PPB).

AI fairness is about people, not just technology.

Questions

